# AIOps PLE Platform

## Architecture Design Document

PLE + adaTT-Based Financial Product Recommendation System
Technical Design Document (Internal Reference)

Version 1.0 — 2026-04-01

**Design vs Implementation Note**

This document describes the **design intent and target architecture**. The current implementation may be a subset of the design, and some components described herein may not yet be implemented. Refer to the codebase and `pipeline_state.json` for implementation status.

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# 1. System Overview

## 1.1. System Purpose

AIOps PLE Platform is an end-to-end multi-task learning platform for **financial product recommendation**. It targets banks, card companies, and financial holding companies, generating personalized recommendations for comprehensive financial products—deposits, cards, loans, investments—along with **explainable rationale** simultaneously.

The final deliverable of AI recommendation is not a probability (0.73) but a **reason the customer can accept**. The persuasion target is always a person, and three levels of explanation are required:

| Audience | Question | Expected Level |
|---|---|---|
| Customer | "Why this product?" | Trust -> Conversion |
| Banker | "Why recommend this to this customer?" | Sales rationale |
| Regulator | "Why was this decision made?" | Regulatory compliance (EU AI Act, FSS Guidelines) |

## 1.2. Target Users and Operating Environment

- **Financial ML Operations Staff**: Realistically 1–2 people. N models x explanation modules x ensemble logic = N management points. A single config system must enable unified management.
- **Hardware**: 1–4 GPU scale. Since MLP parameter scaling is infeasible, structural inductive bias is used to secure expressiveness.
- **Infrastructure**: On-Prem (Airflow + DuckDB) or AWS (SageMaker + S3). The architectural philosophy remains identical; only the infrastructure layer changes.

## 1.3. On-Prem vs AWS Comparison

| Aspect | On-Prem (Current) | AWS (Target) | Migration Rationale |
|---|---|---|---|
| Orchestration | Airflow 86 DAGs ($300/mo) | Step Functions 5 ($0/mo) | Pay-per-execution |
| Training | Local GPU | SageMaker Spot (70% savings) | Parallel ablation |
| Storage | DuckDB files | S3 Parquet | Durability, IAM |
| Serving | FastAPI + Docker | Lambda / ECS Fargate | Auto-switch by scale |
| Experiment Mgmt | MLflow (Docker) | SageMaker Experiments | Eliminate server maintenance cost |
| Monitoring | Custom drift_monitor | SageMaker Model Monitor | Managed service |

# 2. Design Philosophy

Four core design principles serve as the starting point for all technical decisions.

## 2.1. Principle 1: Inherent Explainability

The system pursues **structural explanation** rather than post-hoc explanation (SHAP/LIME).

- **SHAP/LIME limitations**: Separated from the model, so explanations diverge from internal behavior. Explanations change drastically with slight input changes. Separate computation per inference multiplies serving latency.
- **Structural approach**: Explanations are derived from the model structure itself (gate, evidential, contrastive). A single forward pass produces both inference and explanation simultaneously.

The Heterogeneous Expert architecture makes this possible:

- CGC gate weights themselves serve as explanations: "Temporal(0.35), DeepFM(0.28), HGCN(0.22)"
- Expert names directly map to business context: "Time-series pattern contributed 35%, feature interaction 28%, product hierarchy 22%"
- With homogeneous MLP experts, "MLP 3 contributed" carries no business meaning

## 2.2. Principle 2: Graceful Degradation

Even if one expert becomes useless, the remaining experts naturally compensate through gate redistribution.

- Ablation proves "removing any expert does not cause catastrophic performance degradation"
- Financial sector characteristics: Unlike big tech's aggressive "AUC +0.01 = N billion in revenue" experimentation, financial firms face "regulatory sanctions for model malfunction" -> conservative operations, stability first

## 2.3. Principle 3: Extensibility

Adding new features/tasks requires only config changes. adaTT automatically learns new relationships without modifying existing structures.

- **Pool/Basket/Runtime 3-tier**: Code(Pool) -> Config(Basket) -> Training(Runtime) separation
- Zero code changes for domain switching: simply replace config files from `configs/financial/` to `configs/ecommerce/`

## 2.4. Principle 4: Manageability

The entire pipeline (feature generation -> training -> distillation -> serving -> recommendation rationale) is managed through a unified config system (`pipeline.yaml` + `feature_groups.yaml`).

- **Config-Driven**: A single YAML defines data/tasks/model/infrastructure. New problems without code changes.
- **Registry Pattern**: Expert, Task, Feature, Model, Tower are all registered via plug-in pattern.
- **Schema-First**: The data schema determines the entire pipeline.

# 3. Architecture Decision History

## 3.1. From ALS to Black-Litterman

The existing financial product recommendation system was based on ALS (Alternating Least Squares) collaborative filtering. When the decision to adopt MLOps was made, next-generation model evaluation began.

Initial candidates:
1. DL model family (DeepFM, Wide&Deep, AutoInt)
2. GBM model family (XGBoost, LightGBM, CatBoost)
3. DL + GBM ensemble

During ensemble evaluation, there was an attempt to apply the financial portfolio optimization **Black-Litterman model** to recommendations. The idea was to treat each model's prediction as an "expert view" and integrate them via Bayesian updating weighted by uncertainty (risk).

## 3.2. Black-Litterman Drop Rationale

Critical limitations were identified at the design stage:

| Limitation | Details |
|---|---|
| Business interpretation impossible | Each model's contribution is blended during Bayesian updating -> "Why was this product recommended?" becomes hard to explain. Fatal in the regulatory environment. |
| Structural mismatch | Fundamental difference between financial portfolios (continuous weight allocation) and product recommendation (discrete selection) |
| View matrix automation difficulty | Uncertainty estimation for each model is subjective and hard to automate |
| Multi-task limitation | Insufficient means to express inter-task relationships when integrating churn/recommendation/segment/value into a single BL framework |

## 3.3. PLE + adaTT Selection Process

The question was reframed: "How to blend multiple models well?" (ensemble) -> "Is there a structure where multiple experts collaborate within a single model?" (MoE/PLE).

MTL (Multi-Task Learning) family evaluation:

| Architecture | Characteristics | Limitation |
|---|---|---|
| Shared-Bottom | Single expert shared across all tasks | Severe negative transfer |
| MMoE | N experts + per-task gate | All experts exposed to all tasks -> insufficient specialization |
| **PLE** | Shared + task-specific expert separation, CGC gate | Selected |

**Why PLE was selected:**
- Expert network = internal ensemble (per-expert specialization)
- CGC gate = per-task expert weighting -> explainability
- Single model training/deployment -> 1 management point, fixed serving cost
- What Black-Litterman tried externally (uncertainty-based integration of expert opinions) is solved internally, data-driven, within the model structure

## 3.4. Decision Flow Summary

```
ALS (existing)
  | "MLOps adoption, next-gen model needed"
DL + GBM ensemble evaluation
  | "Simple ensemble insufficient for multi-task integration"
Black-Litterman attempt
  | "Financial portfolio != product recommendation, dropped at design stage"
PLE + adaTT selected
  | "Prototype validated on-premises"
AWS migration
  | "Parallel ablation + end-to-end serving needed"
Benchmark data + Ablation execution
```

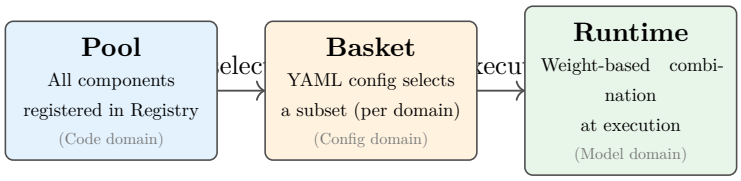


Figure 1: Pool / Basket / Runtime 3-tier architecture.

```
  | (planned)
Distillation + Lambda serving
```

# 4. PLE + Heterogeneous Expert Basket

## 4.1. Limitations of Original PLE and Introduction of Heterogeneous Experts

The original PLE (Tencent, 2020) shared experts consist of **homogeneous MLP x N** (same structure, different initialization). Expressiveness increases only by adding parameters, so lightweight models lack representational power.

Our approach: shared expert = **7 heterogeneous experts**. Each expert processes the same data from a different perspective with a different inductive bias, and the CGC gate learns "which perspective is useful for this task." Lighter than a single large MLP yet more expressive.

## 4.2. Pool / Basket / Runtime 3-Tier

| Tier | Role | Change Agent |
|---|---|---|
| Pool | Register all available components in Registry | Developer (code addition) |
| Basket | Select subset for a specific pipeline via YAML | Operator (config swap) |
| Runtime | Weighted combination of selected components during execution | Model (automatic during training) |

## 4.3. Expert Pool — 11 Registered Types

| # | Registry Name | Class | Primary Axis | Basket |
|---|---|---|---|---|
| 1 | `deepfm` | DeepFM Expert | State (feature interaction) | O |
| 2 | `temporal_ensemble` | Temporal Ensemble | Timeseries (Mamba+LNN+Transformer) | O |
| 3 | `hgcn` | Unified HGCN | Hierarchy (hyperbolic graph) | O |
| 4 | `perslay` | PersLay Expert | Snapshot (TDA global) | O |
| 5 | `causal` | Causal Expert | Snapshot (NOTEARS DAG) | O |
| 6 | `lightgcn` | LightGCN Expert | Item (graph collaborative filtering) | O |
| 7 | `optimal_transport` | OT Expert | Snapshot (Sinkhorn) | O |
| 8 | `mlp` | MLP Expert | State (basic) | — |
| 9 | `mamba` | Mamba Expert | Timeseries (SSM) | — |
| 10 | `autoint` | AutoInt Expert | State (Self-Attention) | — |
| 11 | `xdeepfm` | XDeepFM Expert | State (CIN + Deep) | — |

## 4.4. 7-Expert Selection Rationale

Each expert possesses a non-overlapping inductive bias:

| Expert | Inductive Bias | Pattern Captured with Fewer Parameters |
|---|---|---|
| DeepFM | Feature interaction | 2nd-order cross effects between features |

| Expert | Inductive Bias | Pattern Captured with Fewer Parameters |
|---|---|---|
| Temporal Ensemble | Composite time series (Mamba+LNN+Transformer) | Long/short-term/discontinuous time-series integration |
| HGCN | Hierarchical structure (hyperbolic space) | Product category tree |
| PersLay | Topological structure | TDA persistence shapes |
| LightGCN | Graph relationships | Customer-product collaborative filtering |
| Causal | Causal relationships | Causal direction (DAG constraint) |
| Optimal Transport | Distribution matching | Distribution differences across customer segments |

### 4.4.1. Temporal Ensemble: Intra-Expert Ensemble

Financial time series have compound structures that no single model can capture. The Temporal Expert itself is designed as an internal ensemble of 3 models:

| Model | Strength | Weakness |
|---|---|---|
| Mamba (SSM) | Long-range dependency, O(n) efficiency | Weak on irregular intervals |
| LNN (Liquid NN) | Adapts to discontinuous/irregular data | Poor at long-context summarization |
| Transformer | Short-context extraction | $O(n^2)$, inefficient for long sequences |

## 4.5. CGC Layer + Attention

`CGCLayer` is the core building block of PLE. For each task, a gating network combines shared + task-specific expert outputs.

- **dim_normalize=True**: When expert output dimensions differ, `sqrt(mean_dim/dim)` scaling corrects dimensional imbalance
- **bias_high/bias_low**: Injects initial bias toward domain-relevant experts
- **entropy regularization**: Prevents expert collapse

Stacked PLE: 3 CGC layers. Layer 0 uses the heterogeneous Expert Basket + FeatureRouter (each expert receives a different input_dim subset), while Layers 1–2 use homogeneous MLP experts for abstraction. Because per-expert input dims are heterogeneous (27D–168D), `dim_normalize` corrects dimensional imbalance in the CGC gate.

## 4.6. Dual-Registry Architecture

```
Expert Pool Registry (core.model.experts.registry.ExpertRegistry)
    +-- AbstractExpert(input_dim, config)
    +-- 11 types registered

Expert PLE Registry (core.model.ple.experts.ExpertRegistry)
    +-- BaseExpert(input_dim, output_dim, dropout)
    +-- For CGCLayer default expert creation

Expert Basket (core.model.ple.experts.ExpertBasket)
    +-- Pool Registry -> Basket selection -> Inject into CGCLayer.shared_experts
```

# 5. adaTT + Task Group

## 5.1. 4 Financial DNA Groups

The initial design considered per-GMM-cluster task sub-heads, but with K=20, T=13, this would produce 260 sub-heads -> unmanageable, high overfitting risk. The direction was shifted to **grouping tasks by financial DNA perspective**:

| Group | Financial DNA | Included Tasks | intra | inter |
|---|---|---|---|---|
| engagement | Does the customer respond? | next_mcc, top_mcc_shift | 0.8 | 0.3 |
| lifecycle | Where is the customer? | churn_signal, product_stability, segment_prediction | 0.7 | 0.3 |
| value | How valuable is the customer? | mcc_diversity_trend | 0.6 | 0.3 |
| consumption | What will the customer buy? | nba_primary, cross_sell_count, will_acquire_* (5) | 0.7 | 0.3 |

A total of **13 tasks** organized into 4 semantic groups.

## 5.2. Adaptive Task Transfer (adaTT)

- **Intra-group**: Strong transfer between tasks in the same group (0.6–0.8)
- **Inter-group**: Weak transfer between different groups (0.3) — minimizing interference
- **Negative transfer threshold**: Automatically blocks transfer when performance degrades
- **EMA decay**: Stabilizes transfer weights
- **Warmup/freeze epochs**: Initial stabilization

## 5.3. Loss-Level Transfer: Logit Transfer (3-Method Dispatch)

Explicit causal relationships between tasks are modeled through 3 edges:

| Source | Target | Method | Meaning |
|---|---|---|---|
| has_nba | nba_primary | output_concat | Subscription status -> Which product |
| churn_signal | product_stability | output_concat | Churn -> Product stability |
| next_mcc | nba_primary | hidden_concat | Next merchant category -> Next product (feature sharing) |

Three transfer methods:

- **residual**: source output -> Linear -> tower_dim, residual addition
- **output_concat**: source output concatenated with tower input -> Linear -> tower_dim
- **hidden_concat**: source pre-tower hidden concatenated with tower input -> Linear -> tower_dim

`logit_transfer_strength: 0.5` — transfer ratio.

## 5.4. HMM Triple-Mode Projection

Three HMM modes are routed to task groups:

| Task Group | HMM Mode | Meaning |
|---|---|---|
| engagement | behavior | Behavior mode -> Response/activity |
| lifecycle | lifecycle | Lifecycle mode -> Churn/retention |
| value | journey | Journey mode -> Value/spending |
| consumption | journey | Journey mode -> Consumption pattern |

Per-mode 16D -> `task_expert_output_dim` projection, followed by additive fusion into tower input.

## 5.5. Multidisciplinary Per-Task Routing

24D multidisciplinary features are routed as 6D slices to 4 task groups:

| Task Group | Discipline | Dimension |
|---|---|---|
| engagement | chemical_kinetics | [0:6] |
| lifecycle | epidemic_diffusion | [6:12] |
| value | crime_pattern | [12:18] |

| Task Group | Discipline | Dimension |
|---|---|---|
| consumption | interference | [18:24] |

# 6. Feature Pipeline

## 6.1. 5-Axis Feature Classification

All features are classified along 5 axes, and each axis is mapped to a corresponding Feature Generator and Expert.

| Axis | Temporal Dependency | Rate of Change | Representative Data | Processing Method |
|---|---|---|---|---|
| State | None (static) | Nearly invariant | Age, gender, enrollment date | Feature interaction (FM) |
| Snapshot | Long-term (monthly/quarterly) | Slow | 12-month transaction topology, HMM states | Long-term pattern extraction |
| Timeseries | Short-term (daily/weekly) | Fast | Recent 90-day transaction sequence | Sequence modeling (SSM) |
| Hierarchy | None (structural) | Slow | MCC code hierarchy, product categories | Hyperbolic embedding |
| Item | None (relational) | Medium | Customer-product interactions | Graph collaborative filtering |

## 6.2. 12 Feature Groups ( 349D Input / 403D after Phase 0)

The raw input tensor is 349D (12 feature groups). After Phase 0 (3-stage normalization, log1p copies appended), the expanded tensor is 403D. With FeatureRouter active, each expert receives a designated subset of this tensor rather than the full 403D. Per-expert input dims: deepfm=168D, temporal_ensemble=139D, hgcn=27D, perslay=32D, causal=161D, lightgcn=100D, optimal_transport=127D. Feature group-to-expert routing is group-level, auto-built at `build_model()` time from `target_experts` declarations in `feature_groups.yaml` — no hard-coded column routing.

### 6.2.1. 4 Base Groups (transform type)

Generated by transforming existing raw columns. No Generator required.

- `base_rfm`: RFM-based features (quantile transform)
- `base_demographics`: Demographic features
- `base_product`: Product holding status
- `base_activity`: Activity metrics

### 6.2.2. 8 Generated Groups (generate type)

Dedicated Feature Generators produce features.

| Group | Generator | Output D | Description |
|---|---|---|---|
| tda_topology | tda_extractor | 70D | Topological analysis based on Persistence Diagrams |
| hmm_states | hmm_triple_mode | 48D | Journey/lifecycle/behavior state estimation |
| hyperbolic_embedding | hyperbolic_embedding | 20D | Hyperbolic space hierarchical structure embedding |
| temporal_pattern | temporal_pattern | variable | Time-series aggregation + periodic encoding |
| multidisciplinary | multidisciplinary | 24D | Chemical kinetics, epidemic diffusion, interference, crime patterns |
| gmm_clustering | gmm | variable | Soft posterior probabilities |
| economics | economics_extractor | 17D | MPC, income elasticity, permanent income |
| model_derived | model_feature_extractor | 27D | Leveraging prior model outputs |

## 6.3. 11 Academic Disciplines (Multidisciplinary Feature Design)

| Discipline | Adopted Element | Meaning for Financial Customers |
|---|---|---|
| Topology | TDA Persistent Homology | Structural shape of consumption patterns |
| Hyperbolic Geometry | Hyperbolic GCN | Distortion-free embedding of product hierarchy |
| Stochastic Processes | HMM State Transitions | Customer lifecycle stage tracking |
| Control Theory | Mamba (State Space Model) | Long-range behavioral dependency |
| Economics | Permanent/transitory income, marginal utility | Structural decomposition of spending capacity |
| Financial Engineering | Risk metrics, Bandit explore/exploit | Product exploration propensity |
| Graph Theory | LightGCN | Behavioral transfer among similar customer groups |
| Statistics | GMM Clustering | Soft segmentation |
| Causal Inference | Causal DAG (NOTEARS) | Causal direction between behaviors |
| Optimal Transport | Sinkhorn Optimal Transport | Distribution shift across segments |
| Neural ODE | Liquid Neural Network | Adaptation to irregular time intervals |

Key insight: These perspectives are non-overlapping. Topology's "shape" and economics' "utility" are entirely different aspects of the same customer, each contributing independently (proven by ablation).

## 6.4. Dual Role of Features: Training + Recommendation Rationale

The value of multidisciplinary features is not just training performance (AUC contribution). Even if removing TDA features only drops AUC by 0.01, the explanation "Your consumption pattern has been consistently showing a stable shape" is impossible to generate without TDA.

**Business context reverse mapping** is built for all features:

- `hmm_lifecycle_prob_growing` -> "Growth-stage customer"
- `mamba_temporal_d3` -> "Spending increase trend over the last 3 months"
- `hgcn_hierarchy_d5` -> "Position close to investment product category"

## 6.5. 3-Stage Normalization Pipeline

```
Stage 1: Power-law detection (skew+kurt -> log-log R^2) + log1p copy generation
Stage 2: StandardScaler (continuous columns only, binary excluded, TRAIN fit only)
Stage 3: Power-law _log copies are not scaled (raw magnitude preserved)
```

- Scaler is **fit on TRAIN split only**. val/test are transform-only with the train-fitted scaler.
- Binary columns are excluded from scaling.

# 7. Training Pipeline

## 7.1. Phase 0: Data Preparation

A 10+ stage PipelineRunner transforms raw data into training-ready tensors.

| Stage | Name | Description | GPU |
|---|---|---|---|
| 1 | DataAdapter | DuckDB-native raw data load, AdapterRegistry-based | — |
| 1.5 | TemporalPrep | Sequence truncation (drop last month), prod_* recalculation | — |
| 2 | SchemaClassifier | Classify all columns into 5-Axis | — |
| 3 | EncryptionPipeline | PII -> SHA256 -> INT32 (domain-specific salt) | — |
| 4 | FeatureGroupPipeline | Run 8 Generators + PowerLawAwareScaler | cuDF |
| 5 | LabelDeriver | Config-driven 14-label generation | — |
| 5.5 | LeakageValidator | Sequence/correlation/product/temporal 4-way leakage check | — |
| 6 | SequenceBuilder | flat -> 3D tensor (event_seq, session_seq) | — |

Phase 0 runs on **CPU instances**. GPU instances are not wasted on Phase 0.

### 7.1.1. 4-Layer Data Leakage Prevention

1. **Sequence truncation**: 17 months -> 16 months (drop last month)
2. **Product recalculation**: Recalculate prod_* based on month 16 state
3. **Temporal split**: temporal split + gap_days (minimum 7 days)
4. **LeakageValidator**: Sequence/correlation/product/temporal verification, auto-remove features with >0.95 correlation

### 7.1.2. Data Processing Backend Policy

Priority: cuDF (GPU) -> DuckDB (CPU columnar) -> pandas (last resort fallback, only for fewer than 10K records). Direct use of `pd.read_parquet()`, `pd.concat()`, `df.apply()` and similar pandas calls is avoided.

## 7.2. Phase 1–3: Ablation Study (48 Scenarios)

Systematic experiments across 4 dimensions x multiple scenarios:

| Phase | Dimension | Content | Job Count |
|---|---|---|---|
| 1 | Feature Group | full + base_only + bottom-up + top-down | 16 |
| 2 | Expert | deepfm baseline + bottom-up + top-down + mlp_only | 16 |
| 3 | Task x Structure | 4 tiers x 4 structures (shared_bottom / ple_only / adatt_only / full) | 16 |

- **Bottom-up**: base + X -> measures independent contribution
- **Top-down**: full − X -> measures irreplaceability

## 7.3. Model Training (Common to Phase 1–3)

### 7.3.1. Per-Task Loss Dispatch

| Loss Type | Module | Usage | Task Type |
|---|---|---|---|
| focal | FocalLoss(alpha, gamma) | Imbalanced binary classification (calibrated alpha) | binary |
| huber | SmoothL1Loss | Outlier-robust regression | regression |
| mse | MSELoss | Basic regression | regression |
| ce | CrossEntropyLoss(weight) | Multi-class (auto class_weights) | multiclass |
| infonce | InfoNCELoss(temperature) | Contrastive learning | contrastive |

In AMP (Mixed Precision) environments, tower output is cast to FP32 before loss computation (overflow prevention).

### 7.3.2. Uncertainty Weighting (Kendall et al.)

$$L_{\text{total}} = \sum_k \left[ \exp(-s_k) \cdot L_k + \frac{s_k}{2} \right]$$

$s_k$: learnable log-variance for task $k$. Tasks with higher uncertainty -> automatic weight reduction.

### 7.3.3. Evidential Deep Learning (Config-Gated)

| Task Type | Distribution | Parameters | Uncertainty |
|---|---|---|---|
| Binary | Beta(alpha, beta) | alpha, beta | 1/(alpha+beta) |
| Multiclass | Dirichlet(alpha) | alpha_1..K | K/sum(alpha) |
| Regression | Normal-Inverse-Gamma | mu, v, alpha, beta | beta/(v*(alpha-1)) |

### 7.3.4. SAE Regularization (Detached, Config-Gated)

Anthropic-style Sparse Autoencoder. Applied to the shared expert concatenated output. **Detached** so it does not affect main model gradients (analysis-only sidecar). Tied weights, pre-bias centering, ReLU activation.

# 8. Distillation + Serving

## 8.1. Phase 4: Knowledge Distillation (PLE -> LGBM)

```
PLE Teacher (GPU training)
    | Soft label generation (temperature=5.0)
    | Store in S3
LGBM Student (CPU training)
    | alpha=0.3 (30% hard + 70% soft)
    | IG-based feature selection (top-k features)
    | Save lightweight model
Serving (real-time: LGBM ~5ms, batch: PLE)
```

- **Temperature**: 5.0 — maximizes information content of soft labels
- **Alpha**: 0.3 — 30% hard label + 70% soft label blend
- **Fidelity validation**: Teacher-student prediction agreement verification

## 8.2. Phase 5: Serverless Serving (Lambda)

### 8.2.1. Serving Architecture

- **Real-time**: LGBM student -> Lambda ( 5ms latency)
- **Batch**: PLE teacher -> SageMaker Batch Transform
- **Scale switching**: Lambda (default) ECS Fargate (large scale) automatic switching

### 8.2.2. 3-Layer Fallback Architecture (FallbackRouter)

Distillation failures on minority-class tasks motivated a layered serving design. The `FallbackRouter` monitors per-task-type fidelity on a rolling window and escalates when any layer falls below its configured threshold:

| Layer | Model | When Active | Latency |
|---|---|---|---|
| L1 | LGBM student (distilled) | Default: fidelity > threshold on all task types | ~5ms |
| L2a | PLE teacher (direct inference) | LGBM fidelity drops on binary or multiclass tasks | ~80ms |
| L2b | Bedrock LLM (Sonnet) | Regulatory explanation required or confidence < floor | ~800ms |
| L3 | Rule-based fallback | All models unavailable or circuit breaker open | <1ms |

Calibration is maintained separately per layer. L2b (Bedrock) handles the 3-agent reason pipeline (FactExtractor → TemplateEngine → SelfChecker) and is invoked only when explainability depth exceeds what the student's IG attribution can provide. End-to-end serving cost including rationale generation is approximately $0.69 per cycle.

### 8.2.3. FD-TVS Composite Scoring

Task-level predictions are integrated into a single recommendation score:

| Task | Weight |
|---|---|
| nba_primary | 0.30 |
| cross_sell_count | 0.15 |

| Task | Weight |
|---|---|
| churn_signal | 0.15 |
| product_stability | 0.10 |

### 8.2.4. DNA Modifier

Segment-level weight adjustment: TOP(1.3), PARTICULARES(1.0), UNIVERSITARIO(0.8), UNKNOWN(0.7).

### 8.2.5. Constraint Engine

| Constraint | Description |
|---|---|
| Fatigue | Maximum 5 messages within 7 days |
| Eligibility | min_score > 0.05, max_churn_prob < 0.6 |
| Owned Product | Exclude already-owned products via prod_* prefix |
| Product Tier | Minimum tenure: standard 3 months, growth 6 months, premium 12 months |
| Top-K Diversity | Ensure diversity via MMR (lambda=0.5) |

### 8.2.6. Recommendation Rationale Generation Pipeline

```
[Model Inference] -> [IG Attribution: Top Features] -> [Business Reverse Mapping]
-> [LLM Agent: Context Composition -> Natural Language Recommendation Rationale]
```

Four levels of explanation are provided simultaneously:

1. **gate weight**: "Which perspective contributed" (expert-level)
2. **contrastive**: "Why this and not that" (contrastive)
3. **evidential**: "How confident is this" (uncertainty quantification)
4. **SAE**: "Which internal concepts were activated" (neuron-level)

# 9. Monitoring + Compliance

## 9.1. Pipeline Tracking

| Artifact | Location | Purpose |
|---|---|---|
| `pipeline_manifest.json` | output_dir/ | Full pipeline config snapshot |
| `pipeline_state.json` | output_dir/ | Per-stage completion/failure status, resume support |
| `feature_stats.json` | output_dir/ | Zero-variance, NaN ratio, feature column count |
| `label_stats.json` | output_dir/ | Class balance, positive rate |

## 9.2. Per-Stage Checkpoints

| Stage | Artifact | Format |
|---|---|---|
| Feature | `features.parquet` | Parquet |
| Label | `labels.parquet` | Parquet |
| Sequence | `sequences.npy`, `seq_lengths.npy` | NumPy |
| Scaler | `scaler_params.json` | JSON |
| Leakage | `leakage_report.json` | JSON |

## 9.3. Audit Artifacts

```
audit/
+-- schema/          <- Schema validation results
+-- encryption/      <- PII processing audit log
+-- scaler/          <- scaler_params.json
+-- labels/          <- label_transforms.json
+-- leakage/         <- LeakageValidator results
+-- fidelity/        <- Distillation fidelity verification
```

## 9.4. Interpretability Pipeline (Stage 8.5)

| Analysis | Purpose | Output |
|---|---|---|
| Integrated Gradients (IG) | Feature attribution measurement | attribution scores |
| Expert Redundancy CCA | Inter-expert redundancy detection | CCA correlation matrix |
| CGC Gate Analysis | Per-task expert weight analysis | attention heatmap |
| HGCN Interpretable | Hierarchical structure explanation | hierarchy paths |
| Multi Interpreter | Multidisciplinary interpretation integration | structured reasons |
| Template Reason Engine | Natural language recommendation rationale | text templates |
| XAI Quality Evaluator | Explanation quality evaluation | quality scores |
| Model Card | Auto-generated model documentation | model_card.json |

## 9.5. Encryption Pipeline (Stage 3)

```
Schema (pii: true)
    | derive_from_schema()
EncryptionPolicy (per source, per column)
    |
EncryptionPipeline.process_source()
    +-- Step 1: Drop (phone, email, SSN, etc.)
    +-- Step 2: SHA256 Hash (domain-specific salt)
    +-- Step 3: Integer Index (hash -> INT32)
    +-- Step 4: Audit report
```

16 PII domains defined (CUSTOMER, ACCOUNT, CARD, MERCHANT, etc.). SaltManager manages per-domain salts from AWS Secrets Manager or local storage.

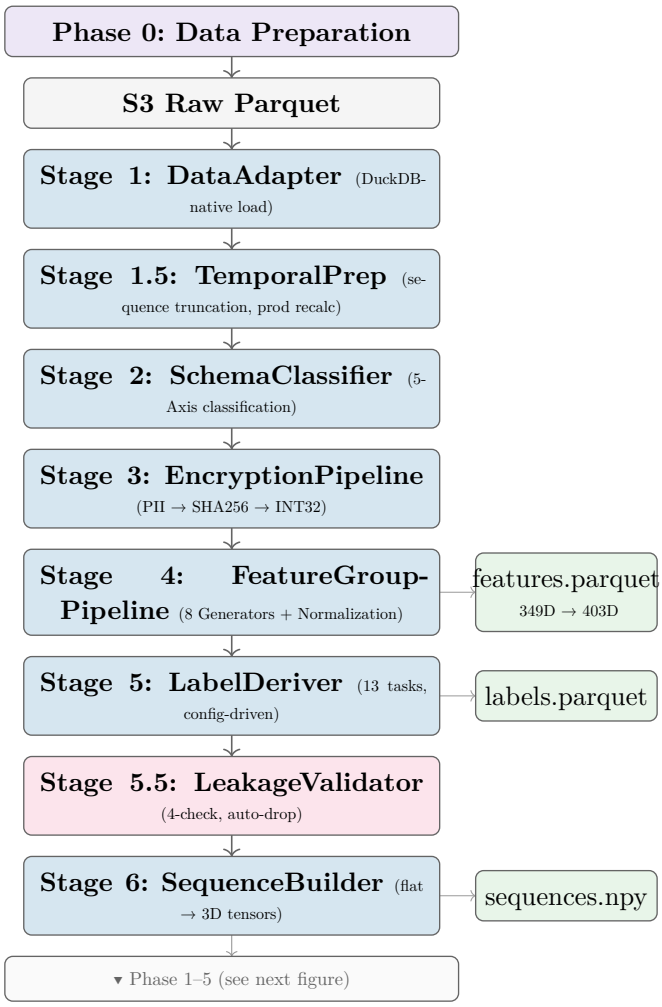


Figure 2: End-to-End pipeline (1/2): Phase 0 data preparation — from raw S3 ingestion to feature/label/sequence tensor storage.

# 10. End-to-End Data Flow

## 10.1. Full Data Flow Diagram

## 10.2. Internal Model Data Flow

FeatureRouter is **active**: each expert receives only its designated feature group subset, not the full 403D tensor ( 349D input; 403D after Phase 0 log1p expansion). Per-expert input dims: deepfm=168D, temporal_ensemble=139D, hgcn=27D, perslay=32D, causal=161D, lightgcn=100D, optimal_transport=127D. Routing is group-level, auto-built from `target_experts` in `feature_groups.yaml`. This reduced model parameters from 4.77M to  2.8M.

## 10.3. GPU/CPU Acceleration Mapping

| Stage | Target | CPU Path | GPU Path |
|---|---|---|---|
| 1 | Data loading | DuckDB (primary) | cuDF (optional) |
| 4 | Generator execution | pandas fallback | cuDF primary (cuML for GMM) |
| 4 | TDA persistence | ripser (NumPy) | cuPY + ripser (5–10x) |
| 4 | StandardScaler | NumPy | cuPY (3–5x on 100M+ rows) |
| 7 | Training data loading | PyArrow (zero-copy) | — |
| 8 | Model training | PyTorch CPU | PyTorch CUDA + AMP |

GPU acceleration is optional; CPU paths are used as automatic fallback when cuDF/cuPY is not installed.

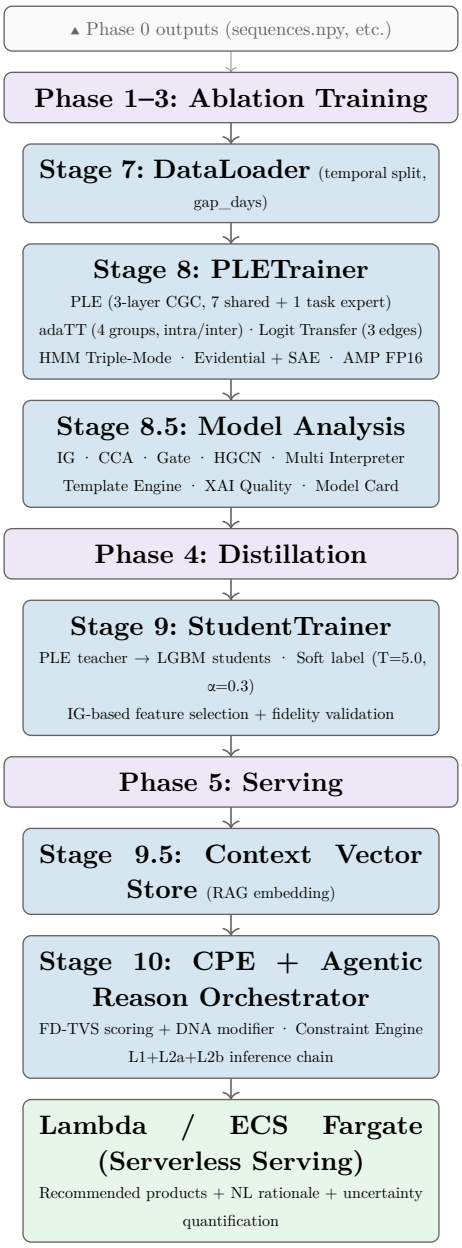


Figure 3: End-to-End pipeline (2/2): Phase 1–3 training → Phase 4 distillation → Phase 5 serving.

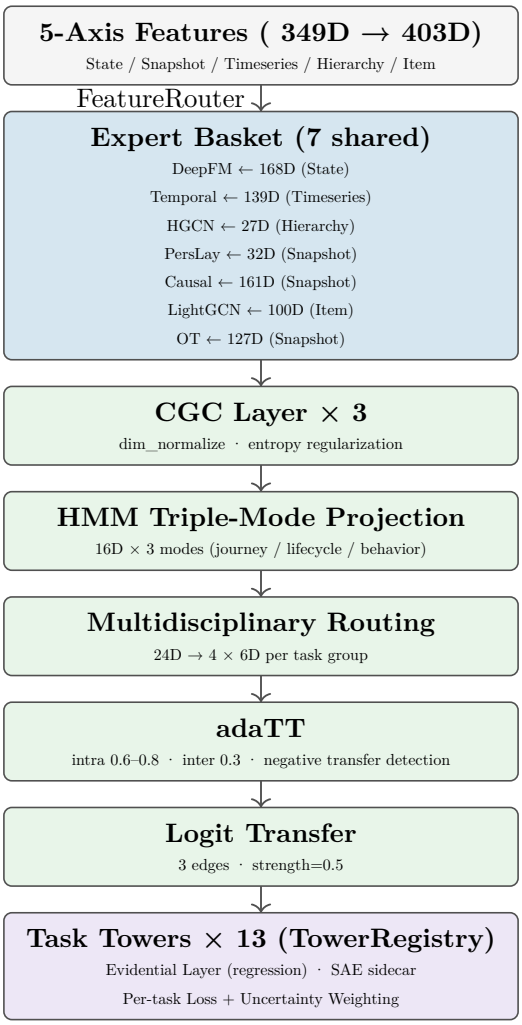


Figure 4: Internal model data flow: FeatureRouter slices per-expert subsets from the 403D feature tensor ( 349D input, 403D after Phase 0 log1p expansion). Routing is group-level, auto-built from `target_experts` in feature_groups.yaml.

# 11. Appendix: PLEModel Build Automation

When `PLEModel.__init__(config: PLEConfig)` is called, the following are built automatically in order:

1. `_build_extraction_layers()` — Stacked CGC + FeatureRouter
2. `_build_cgc_attention()` — Per-task attention (dim_normalize)
3. `_build_task_experts()` — GroupTaskExpertBasket or MLP fallback
4. `_build_hmm_projectors()` — 3 modes x projection
5. `_build_adatt()` — Adaptive Task Transfer
6. `_build_logit_transfer()` — 3-method dispatch, 3 edges
7. `_build_multidisciplinary_routing()` — 24D -> 4 x 6D
8. `_build_task_towers()` — TowerRegistry (standard/contrastive)
9. `_build_evidential_layers()` — NIG for regression (config-gated)
10. `_build_sae()` — Sparse Autoencoder (config-gated)
11. `_build_task_loss_fns()` — `build_loss()` per task
12. `_build_loss_weighting()` — Uncertainty / GradNorm / DWA / Fixed

### 11.0.1. PLEInput Data Container

```python
@dataclass
class PLEInput:
    features: Tensor                      # (batch, input_dim)
    feature_group_ranges: Optional[Dict] # group->(start,end) for routing
    cluster_ids: Optional[Tensor]        # (batch,) cluster assignment
    cluster_probs: Optional[Tensor]      # (batch, n_clusters) soft probs
    targets: Optional[Dict[str, Tensor]] # {task_name: label}
    hyperbolic_features: Optional[Tensor]   # (batch, 20)
    tda_features: Optional[Tensor]          # (batch, 70)
    collaborative_features: Optional[Tensor]# (batch, 64)
    hmm_journey: Optional[Tensor]           # (batch, 16)
    hmm_lifecycle: Optional[Tensor]         # (batch, 16)
    hmm_behavior: Optional[Tensor]          # (batch, 16)
    event_sequences: Optional[Tensor]       # (batch, seq_len, feat_dim)
    multidisciplinary_features: Optional[Tensor]  # (batch, 24)
    sample_weights: Optional[Tensor]        # (batch,)
```

# 12. Appendix: Config File Structure

All system parameters are managed through a split-config scheme. `config_builder.py` is the **single source of truth**: it merges `pipeline.yaml` (infrastructure-agnostic) and `datasets/santander.yaml` (dataset-specific) into the resolved config consumed by all pipeline stages. No pipeline stage reads raw YAML directly.

**`pipeline.yaml`**: Task definitions, model structure, training parameters, AWS infrastructure settings

- `tasks`: 13 tasks (name, type, loss, loss_weight, label_col)
- `model.ple`: num_layers, extraction_dim, expert_basket
- `model.adatt`: task_groups, intra/inter strength
- `model.logit_transfers`: 3 transfer edges
- `training`: batch_size, epochs, learning_rate, amp
- `aws`: instance_type, spot, budget_limit

**`datasets/santander.yaml`**: Dataset-specific overrides (column names, split parameters, adapter settings). Adding a new dataset requires only this file — no changes to `pipeline.yaml` or any Python code.

**`feature_groups.yaml`**: 12 feature group definitions

- `group_type`: transform (existing column transformation) | generate (Generator invocation)
- `generator`: Reference to Pool's Generator name
- `generator_params`: input_filter (dtype, exclude_binary, min_nunique, etc.)
- `target_experts`: List of experts that receive this group's features
- `output_dim`: Output dimension

# 13. Ops/Audit Agent Layer

Two autonomous diagnostic agents operate as a separate layer, asynchronously monitoring the entire pipeline:

- **OpsAgent**: monitors 7 checkpoints (ingestion through A/B testing), performs cross-checkpoint correlation analysis
- **AuditAgent**: audits 5 viewpoints (fairness, concentration, reason quality, regulatory compliance, data lineage)

48 checklist items are automatically evaluated. WARN/FAIL items undergo 3-agent consensus (Sonnet×3) producing diagnostic results with minority report preservation. Diagnostic history accumulates in a LanceDB-based `DiagnosticCaseStore`.

Fully decoupled from the serving path (3 serving agents) — agent failures never degrade customer-facing service.

## 13.1. Serverless Scheduling: EventBridge + Lambda

Both OpsAgent and AuditAgent are deployed as **serverless Lambda functions** scheduled by Amazon EventBridge. This eliminates the need for always-on compute for diagnostic workloads:

- **OpsAgent Lambda**: triggered on a configurable cron (e.g., every 6 hours). Reads pipeline checkpoint artifacts from S3, performs correlation analysis, writes findings to DynamoDB.
- **AuditAgent Lambda**: triggered after each SageMaker training job completion event via EventBridge rule. Reads model outputs and feature stats, performs fairness and lineage checks, writes audit records to DynamoDB.

## 13.2. CloudFormation Monitoring Stack

The monitoring infrastructure is defined as a single CloudFormation stack, ensuring reproducible deployment and version-controlled infrastructure:

| Component | Role |
|---|---|
| EventBridge Rules | Schedule OpsAgent + AuditAgent Lambda; route SageMaker job events |
| CloudWatch Alarms | Drift threshold breach, Lambda error rate, serving latency P99 |
| SNS Topics | Alert routing: WARN → Slack webhook, FAIL → PagerDuty + email |
| DynamoDB Tables | DiagnosticCaseStore (audit history), PipelineState (job resume state) |
| Lambda Functions | OpsAgent, AuditAgent, FallbackRouter health check |

CloudWatch metric filters extract per-task validation metrics from SageMaker training logs and expose them as custom metrics. This enables CloudWatch Alarms to fire on task-level degradation without requiring a separate monitoring service.

Detailed design: Design Document 11 (`docs/design/11_ops_audit_agent.typ`)

## 13.3. On-Premises (Air-Gapped) Deployment

In air-gapped on-premises environments, the same pipeline runs on local GPU (RTX 4070), using Exaone 3.5 7.8B (reason generation) and Qwen 2.5 14B Q4 (agent consensus) instead of Bedrock. Data never leaves the premises, providing structural data protection.

When switching domains, replacing only these 2 files configures an entirely different recommendation system without any code changes.